

Peer-Review Reassessment of the Exoplanet Portfolio

Executive summary

Taken as a portfolio for graduate-school admission, research-assistant applications, or an early-career scientific-data role, this is **credible scientific work and materially stronger than a collection of plotting exercises**. The strongest evidence is not the number of repositories; it is the repeated pattern of taking traceable published data products, doing an independently coded calculation on them, and distinguishing that local calculation from the stronger inference made by the originating paper. Examples include the HAT-P-11b repository explicitly separating its 0.827% weighted-mean helium decrement from Allart et al.'s model-fitted 1.08% measurement; the HD 189733b repository distinguishing local band-contrast signal-to-noise from molecular retrieval significances; K2-18b explicitly refusing to call a hand-selected CO₂-band statistic a retrieval; and LHS 475b stating that a model being statistically consistent is not the same as that model being detected. Those distinctions are exactly the sort of epistemic discipline I look for in a junior researcher. The atmospheric context is also generally current and well sourced: for example, the WASP-107b discussion now correctly attributes the 4.2 σ methane detection and ~1 ppm abundance to JWST/NIRSpec work rather than to a different instrument, in agreement with Sing et al. ¹

The main weakness is that the portfolio sometimes **looks more statistically mature than it actually is**. Most of the planet repositories work on already-reduced spectra rather than detector-level or time-series data, and their independent analyses are usually first-order summaries: inverse-variance weighted means, differences of means, diagonal χ^2 , single-wavelength Planck inversion, or averaging derived brightness temperatures. Those are legitimate analyses, but they do not usually incorporate spectral covariance, shared reduction systematics, correlated phase-bin posteriors, stellar-parameter uncertainty, or full posterior propagation. Several quoted uncertainties therefore answer a narrower question than their visual prominence suggests. The method repositories are good pedagogical injection-recovery demonstrations but mostly use favourable synthetic signals and simplified noise models; they validate the code under its own assumed generative model more than they validate performance on messy astronomical data.

My admissions-committee interpretation would therefore be: **strong evidence of scientific literacy, quantitative competence, self-directed learning and honest technical communication; moderate evidence of research-level statistical inference; limited evidence of end-to-end observational reduction or production-grade scientific software engineering**. For an exoplanet PhD or RA application, I would regard the portfolio favourably provided it is described as *reproducible archival secondary-data analysis and method demonstrations*, not as fifteen pieces of original publishable research. For a data-science job it demonstrates useful Python/Numpy/SciPy competence, but the absence of an evident automated test suite, continuous integration, locked environments, structured APIs and validation against independent implementations keeps it below a strong software-engineering portfolio. The remaining problems are fixable, but several are specialist-visible enough that I would correct them before sending the links to an exoplanet atmospheric physicist.

Verification of the claimed fixes

I treated the supplied checklist of claimed revisions as a separate verification task rather than assuming the fixes worked as intended. filecite~~turn0file0~~ The result is encouraging: **twelve of the fifteen fixes are substantively confirmed, three are only partial, and none of the original issues appears wholly unaddressed**. A few fixes have, however, exposed new problems adjacent to the corrected one.

Claimed revision	Status	Peer-review finding
HAT-P-11b: identify CARMENES correctly and separate local 0.827% statistic from published 1.08% result	Confirmed	README, webpage and code now consistently identify the Allart et al. CARMENES data and explicitly state that the two numbers use different estimators. Published values are consistent with the literature. ²
55 Cnc e: replace incorrectly paired eclipse depths with actual 2012/2013 Demory season values	Confirmed, with a new explanatory error	The code now uses 47 ± 21 and 176 ± 28 ppm and obtains the appropriate $\sim 3.7\sigma$ two-season contrast. Demory et al. report exactly those seasonal depths and a constant-depth rejection at 3.7σ . ³ The new statement attributing the brightness-temperature discrepancy partly to reflected light is weak; see below.
Microlensing: make perturbation-window discovery blind to injection truth	Confirmed	<code>TRUE_*</code> parameters generate the synthetic data but are not passed into the window-finding or fitted models. The larger problem is now the model-matched recovery of q , not leakage of the injected event time.
HD 189733b: stop presenting local band S/N as molecular detection significance	Confirmed	README/code carefully call the 22–25 σ quantities band-contrast S/N and quote the paper's actual retrieval significances separately. Fu et al. report H ₂ O 13.4 σ , CO ₂ 11.2 σ , CO 5 σ and H ₂ S 4.5 σ . ⁴
TRAPPIST-1e: replace inappropriate spectrum with the stellar-decontaminated 2025 JWST analysis	Confirmed, but a new instrument-resolution error appears	The repository now works from the Espinoza et al. NIRSpec/PRISM analysis and appropriately treats a flat spectrum as a constraint rather than an atmospheric detection. Espinoza et al. indeed use four PRISM transmission spectra and rule out cloudy primary H ₂ -dominated atmospheres at $>3\sigma$. ⁵
Direct imaging: normalise contrast to stellar aperture flux and implement Mawet small-sample correction	Partial	Flux normalisation is now physically meaningful and a Student- t correction is implemented. However, the code uses essentially a fixed set of 12 azimuthal apertures at every separation, so its claimed number of independent resolution elements does not scale correctly with annular circumference. Mawet et al.'s correction is explicitly about the separation-dependent number of independent samples. ⁶

Claimed revision	Status	Peer-review finding
K2-18b: use revised data/source and stop treating CO₂ window contrast as a retrieval	Confirmed	The local statistic is explicitly called descriptive and gives $\sim 0.2\sigma$; the README correctly separates it from the independent atmospheric retrieval literature. Schmidt et al.'s extensive reanalysis finds CH ₄ at $\sim 4\sigma$ but no statistically reliable CO ₂ or DMS. ⁷
GJ 1214b: remove brightness-temperature clipping and propagate asymmetric errors	Partial	The obvious point clipping has been improved, but the lower flux perturbation is still forced through <code>max(flux - lower_error, 1e-12)</code> before Planck inversion. Several low-S/N nightside bins can cross zero, so this remains an artificial truncation of the transformed posterior.
WASP-121b: propagate point uncertainties and add uncertainty to day/night contrast	Confirmed narrowly; major conceptual issue remains	The code now propagates per-point errors and constructs a contrast uncertainty. But inverse-variance averaging hundreds of wavelength-dependent brightness temperatures into one ultra-precise scalar temperature is not physically equivalent to estimating a bolometric hemisphere temperature.
HD 209458b: explain that the four reductions share observations/processing and are not a formal systematic error distribution	Partial	Public prose has improved, but the analysis script still describes the four tree leaves as a “genuine reduction-uncertainty band” and “a real estimate of pipeline systematic uncertainty”. The upstream release calls them four <i>leaf spectra</i> from a rule-based reduction tree, not four statistically independent pipelines. ⁸
Transit demo: search duration rather than supplying it, and label S/N as trial-uncorrected	Confirmed	Six duration fractions are searched and the README/code explicitly disclaim a false-alarm interpretation of the 77.8σ depth S/N.
RV demo: correct stellar-mass scaling and constrain fit parameters in the optimiser	Confirmed	The $M_{\star}^{-2/3}$ limiting dependence is now correct and eccentricity, period and semi-amplitude are bounded directly in <code>curve_fit</code> .
LHS 475b: use “reduced χ^2” correctly and identify $\chi^2_{\nu} > 2$ as a heuristic	Confirmed	Both README and webpage now make the distinction explicit.
WASP-96b: acknowledge covariance between morning/evening spectra and make peak-to-trough descriptive	Confirmed	The limitation is clearly disclosed and the statistic is no longer presented as a formal independent-limb significance.

Claimed revision	Status	Peer-review finding
WASP-107b: correct methane attribution/instrument	Confirmed	The repository now distinguishes the NIRISS spectrum it actually analyses from the NIRSpec methane result cited from the literature. Sing et al. report CH ₄ at 4.2σ and 1.0 ± 0.5 ppm. ¹

The pattern here is important: the author is clearly capable of responding intelligently to scientific criticism. That itself is positive evidence for research potential. The places still marked “Partial” are not merely wording preferences; they correspond to real statistical or provenance problems that should be fixed.

Comparative scientific and reproducibility audit

Planet-report repositories

Repository	Data and instrument	Independent analysis actually performed	Uncertainty treatment	Reproducibility judgement
WASP-107b	Published JWST/ NIRISS SOSS spectrum; literature context also cites NIRSpec/MIRI.	Concatenation/ handling of SOSS orders; inverse-variance mean; scatter and peak-to-trough amplitude.	Per-bin quoted errors enter weighted mean; no covariance; max-min has no formal error.	Good for descriptive analysis; not retrieval-grade.
WASP-96b	JWST NIRISS/ SOSS plus NIRSpec/G395H terminator products.	Evening-minus-morning spectrum, weighted average difference, peak-to-trough descriptive comparison.	Quadrature assumes independent limb estimates; covariance explicitly admitted.	Good and honest, but quoted diagonal significance remains first-order.
K2-18b	Combined NIRISS/NIRSpec/ MIRI published spectrum.	CO ₂ -window minus local-continuum weighted means.	Standard errors from published bins; correlation ignored and disclosed.	Scientifically modest but appropriately labelled.
HD 209458b	JWST/MIRI LRS, four leaves of Chubb & Grant's rule-based reduction tree. ⁸	Mean spectrum and spread among reduction leaves.	Photon errors plus empirical leaf spread, but the latter is not a calibrated uncertainty distribution.	Reproducible calculation; interpretation of reduction spread needs repair.

Repository	Data and instrument	Independent analysis actually performed	Uncertainty treatment	Reproducibility judgement
TRAPPIST-1e	Four JWST/NIRSpec PRISM spectra distilled into a decontaminated published spectrum.	Weighted flat-line fit and χ^2 v.	Diagonal errors; no covariance/GP posterior propagated into local χ^2 .	Clear and useful as a first diagnostic.
WASP-121b	JWST/NIRSpec/G395H spectroscopic phase curve.	Day/night phase selection, wavelength-bin brightness-temperature inversion and weighted contrasts.	Per-point asymmetric errors are symmetrised; wavelength/phase covariance omitted.	Technically non-trivial, but headline aggregate uncertainty is physically over-precise.
GJ 1214b	JWST/MIRI phase-curve spectral products from Kempton et al.	Planck inversion by wavelength, day/night contrast, inverse-variance averaging.	Asymmetric source errors reduced to transformed symmetric errors; lower flux crossing zero is still clipped.	Substantive calculation, but uncertainty transform is not trustworthy in low-S/N bins.
HD 189733b	JWST/NIRCam transmission spectrum from Fu et al.	Molecular-band/continuum weighted-mean contrasts.	Quadrature of band means; no spectral covariance or window-selection penalty.	Good independent sanity check, explicitly not a retrieval.
55 Cnc e	Spitzer/IRAC 4.5 μ m season-split occultation depths from Demory et al.	Two-point depth significance and monochromatic Planck brightness-temperature inversion.	Depth errors propagated by $\pm 1\sigma$ transformation; reasonable for the simple calculation.	One of the cleaner demonstrations, after correcting source values.
HAT-P-11b	CARMENES He I 10830 Å reduced light curve and spectral profile.	Weighted in/out-of-transit comparison and line-profile visualisation.	Treats phase-folded points as independent; limitation explicitly stated.	A genuine independent measurement, but not a substitute for the transit model.

Repository	Data and instrument	Independent analysis actually performed	Uncertainty treatment	Reproducibility judgement
LHS 475b	JWST/NIRSpec G395H spectrum and published atmosphere models.	Weighted flat fit and χ^2 comparisons to four forward models.	Diagonal Gaussian χ^2 ; no covariance; model offset handled upstream rather than fitted locally.	A useful model-comparison exercise; statistical calibration could be one step more rigorous.

The external literature generally supports the broad scientific narratives. LHS 475b's featureless G395H spectrum rules out a primordial H_2 -dominated atmosphere and a cloud-free pure-methane case but remains compatible with high cloud, a tenuous atmosphere or no appreciable atmosphere. ⁹ GJ 1214b's published JWST phase-curve analysis finds average brightness temperatures of 553 ± 9 K and 437 ± 19 K and favours a high-metallicity cloudy/hazy atmosphere. ¹⁰ HD 189733b's molecular detections are retrieval results, not merely visually chosen spectral bands, exactly as the repository now acknowledges. ⁴

One distinction should be made very clear on the portfolio landing page: **“uses real archival data” does not mean “performs the original reduction from raw telescope data.”** These repositories mostly operate on published, reduced spectra or secondary products from Zenodo/MAST-associated releases. That is absolutely legitimate work, and for a student portfolio it is valuable. But a specialist will distinguish it immediately from calibrating JWST ramps, modelling instrument systematics, fitting transit light curves, carrying reduction choices into posteriors and running atmospheric retrievals. Chubb & Grant's HD 209458b work, for example, explicitly develops a tree-structured reduction so uncertain reduction decisions can enter subsequent modelling; the portfolio begins downstream of most of that work. ¹¹

Detection-method repositories

Repository	Core implementation	Injection-recovery strength	Statistical realism	Software/reproducibility judgement
Transit photometry	Custom folded-box period/phase/duration search.	Period, duration and depth genuinely searched rather than supplied.	Gaussian white noise, no gaps/red noise; S/N not trial corrected, correctly disclosed.	Clear procedural code and fixed RNG seed; compare formally against <code>astropy.timeseries.BoxLeastSquares</code> and add tests.

Repository	Core implementation	Injection–recovery strength	Statistical realism	Software/reproducibility judgement
Radial velocity	Kepler equation solved from scratch; Keplerian RV model; SciPy Lomb–Scargle and nonlinear fit.	Irregular seasonal sampling; period/K/mass recovered cleanly.	Signal is extremely high S/N; jitter is injected but not inferred; no FAP or model comparison.	Probably strongest method-code repo; numerical tests and posterior inference would materially improve it.
Microlensing	PSPL model, residual anomaly search and Gaussian planetary perturbation approximation.	Perturbation location is found blindly.	Planet signal is not a binary-lens caustic; q is recovered by inverting the same scaling law used to inject it.	Pedagogically useful but significantly more toy-like than README rhetoric implies.
Direct imaging	Synthetic ADI sequence, median PSF subtraction, de-rotation, aperture S/N and contrast curve.	Companion is genuinely rotated through an ADI sequence and recovered after subtraction.	Speckles are simplified; small-sample correction uses an incorrect notion of independent apertures; no throughput map.	Good image-processing demonstration, not yet a scientifically calibrated contrast-curve pipeline.

Code style across the four method repositories is readable: constants are gathered near the top, fixed random seeds make demonstrations repeatable, functions have sensible scientific names, `Path` is used rather than brittle working-directory assumptions, and output statistics are written to machine-readable CSV files. The example dependency file uses broad lower bounds such as `numpy>=1.24`, `scipy>=1.10`, `matplotlib>=3.7` and `scienceplots>=2.1`; that is convenient but not a frozen computational environment. I would trust these scripts as **transparent educational calculations that I can inspect and rerun**, but I would not accept their numerical outputs unreviewed in a paper without unit tests, validation cases, pinned environments and—where appropriate—comparison to established astronomical packages.

Repository-by-repository critique

WASP-107b. Scientifically, this is now a sound small analysis wrapped around a real NIRISS spectrum. The repository no longer confuses its NIRISS-based descriptive work with the later NIRSpec methane detection; the latter really is a 4.2σ CH₄ detection at 1.0 ± 0.5 ppm, depleted by roughly three orders of magnitude relative to equilibrium expectations. ¹¹ The code itself—weighted mean, scatter and peak-to-trough structure—is correct as descriptive statistics. The weakness is depth: a spectrum mean and extrema do not independently establish water, aerosols or chemistry, and max-minus-min is particularly sensitive to noise/outlying bins. I would replace any “signature” language arising from the local calculation with “spectral structure consistent with the published interpretation”. Code quality is clean enough for this purpose, but there is no covariance handling or unit-test machinery. Presentation is strong and, importantly, now scientifically honest.

WASP-96b. The morning/evening comparison is real analysis rather than mere plotting, and the repository deserves credit for explicitly saying that the two limb spectra share observations and pipeline systematics, so quadrature errors are only a first-order approximation. That admission is essential because a quoted limb difference significance is otherwise easy to over-interpret. The broader atmospheric description should remain conservative: current combined JWST/ground-based work robustly sees H₂O, CO₂ and Na and finds super-stellar metallicity in self-consistent models, while aerosol/scattering and terminator asymmetry remain more complicated than a simple “cloud-free benchmark” narrative. ¹² Scientifically this is a useful sanity-check repository, but a domain expert would want the covariance matrix or posterior samples before treating any morning–evening significance as an actual hypothesis test. Web presentation is clear; the limitations paragraph saves the scientific credibility of the headline statistics.

K2-18b. This is one of the best examples of restraint in the set. A narrow CO₂-band-versus-continuum statistic of order 0.2σ is correctly described as a descriptive check, not a molecular retrieval, and the code itself does precisely what the text claims. That matters especially for K2-18b, where modelling choices have generated genuinely divergent conclusions: Schmidt et al.’s systematic reanalysis across 60 data treatments and more than 250 retrievals confirms CH₄ at about 4σ but finds no reliable CO₂ or DMS, whereas another 2025 analysis reported $\sim 3\sigma$ evidence for DMS/DMDS in MIRI data; a joint NIRISS+NIRSpec+MIRI analysis subsequently judged the evidence insufficient. ¹³ The only statistical wording I would change is any categorical statement that covariance necessarily widens an uncertainty; the effect depends on covariance structure. Otherwise, the repository does not overclaim what its simple statistic can decide.

HD 209458b. The underlying idea is excellent for a portfolio: take multiple leaves of a real JWST/MIRI reduction decision tree and quantify how the resulting spectrum shifts. Chubb & Grant explicitly introduced a rule-based tree so uncertain reduction choices can be identified and incorporated into modelling, and the associated Zenodo product really does provide four leaf spectra. ¹⁴ The public-facing caveat has improved, but the code still calls the envelope a “genuine reduction-uncertainty band” and “real estimate of pipeline systematic uncertainty”. That is too strong: four correlated leaves are neither random independent draws from a systematic-error distribution nor four independent pipelines. This internal contradiction is exactly the sort of thing a referee will notice. Call it **reduction-choice sensitivity** everywhere. The analysis is otherwise a useful demonstration of robustness analysis, and the webpage is attractive and scientifically interesting.

TRAPPIST-1e. The data choice is now defensible and current. Espinoza et al. analysed four NIRSpec/PRISM spectra, explicitly confronted severe stellar contamination and used Gaussian-process marginalisation before constraining planetary atmospheres; the repository appropriately presents its

flat-line χ^2 as a limited diagnostic and leaves the stronger atmospheric conclusions to the paper. ⁵ However, the webpage's figure description gives NIRSpec/PRISM as approximately $R \sim 10,000$. That is simply wrong by roughly two orders of magnitude: official STScI documentation gives PRISM a nominal resolving power of about $R \sim 100$ over roughly 0.6–5.3 μm ; NIRSpec's high-resolution gratings are only around $R \sim 2700$. ¹⁵ This is an easy typo to repair but a conspicuous one in front of an instrument scientist. Also, $\chi^2_\nu = 0.65$ is not just “comfortably flat”; it can reflect conservative errors and/or correlations, so I would report χ^2 , degrees of freedom and p-value and mention that the local diagonal calculation does not reproduce the stellar-contamination GP covariance.

WASP-121b. This is probably the most technically ambitious planet script and also the one where the headline uncertainty most needs reinterpretation. It propagates uncertainties through a brightness-temperature calculation, extracts day and night phase regions and generates wavelength-dependent contrasts. That is substantive. The problem is the subsequent inverse-variance collapse of hundreds of wavelength-dependent brightness temperatures into a single value with a few-kelvin uncertainty. Brightness temperature is *intrinsically wavelength dependent*: different channels probe different opacities and pressures. The original NIRSpec broadband results themselves give quite different nightside brightness temperatures, about 926 ± 12 K and 1122 ± 10 K in the two detectors, while a later NIRISS analysis estimated bolometric effective temperatures only after explicitly accounting for the unobserved spectral flux, obtaining $T_{\text{day}} = 2717 \pm 17$ K and $T_{\text{night}} \approx 1562 \pm 19$ K. ¹⁶ Therefore the repository's ultra-precise cross-wavelength weighted mean should be labelled a **weighted mean of monochromatic brightness temperatures over this dataset**, not a physical hemisphere temperature. Cross-wavelength and phase posterior covariance would further weaken the tiny quoted formal error.

GJ 1214b. There is real physics here: inversion of thermal planet/star flux into brightness temperature is materially deeper than plotting the source spectrum. But this is currently the most important uncertainty-propagation defect among the planet reports. For a low-S/N nightside bin where $F - \sigma_- \leq 0$, the code substitutes 10^{-12} before inversion. That imposes a positive lower flux and manufactures a finite lower temperature rather than propagating the actual asymmetric/truncated posterior. The correct fix is to use the underlying flux posterior samples, or at minimum Monte-Carlo draws from an explicitly truncated likelihood and report one-sided intervals where appropriate. The final inverse-variance average of wavelength brightness temperatures also differs substantially from the published phase-curve result: Kempton et al. report average dayside and nightside brightness temperatures 553 ± 9 K and 437 ± 19 K, whereas the repo's simplified statistic is noticeably different. ¹⁰ That does not automatically make the repo calculation “wrong”—the estimators differ—but it makes the distinction scientifically important. The webpage's inference that the data demonstrate atmospheric heat redistribution should be tied to the full phase-curve/GCM analysis, not derived from the local two-number contrast alone.

HD 189733b. This is substantially improved. The local weighted band contrasts are mathematically straightforward and the repository now makes very clear that a ~ 22 – 25σ band S/N is **not** a 22 – 25σ molecular abundance detection. The comparison to Fu et al.'s actual retrieval significances— H_2O 13.4σ , CO_2 11.2σ , CO 5σ , H_2S 4.5σ —is correct. ⁴ The residual statistical weakness is that spectral windows and their continuum definitions are chosen after knowing where molecules absorb, shared continuum bins induce dependencies among contrasts, and real reduced spectra may be correlated. Thus the local S/N is best viewed as a feature-amplitude sanity check. I would also avoid overly specific statements assigning HD 189733b's optical blue appearance uniquely to “silicate haze”; high-altitude scattering haze is well established, but mineralogical specificity requires more evidence. Otherwise this is an effective and well-presented portfolio piece.

55 Cancri e. This repository is now quantitatively much cleaner. Demory et al. actually report season depths of 47 ± 21 ppm in 2012 and 176 ± 28 ppm in 2013, with a constant-depth model fitting poorly at the 3.7σ level, so the repository's direct two-point significance check is well grounded. ³ The monochromatic Planck inversion is also a perfectly legitimate educational calculation provided it is not sold as reproducing the paper's temperature inference. The remaining scientific error is the explanation for the discrepancy: the webpage says the simple temperatures are higher partly because the paper accounts for "possible reflected-light contribution". Demory et al. explicitly say they estimated the $4.5\text{-}\mu\text{m}$ temperature using an **observed infrared stellar spectrum** and a mean planet radius; that is the directly documented methodological difference. ¹⁷ The paper passage does not invoke reflected light in deriving those seasonal temperatures. At $4.5\text{ }\mu\text{m}$ I would simply remove that speculative explanation and say "monochromatic blackbody star versus the paper's observed stellar SED/band treatment and adopted system parameters".

HAT-P-11b. This is one of the portfolio's more convincing demonstrations of genuine data analysis. It loads an actual phase-folded CARMENES helium light curve, defines a reproducible phase window, computes weighted in- and out-of-transit means and carries their errors into a flux decrement. Its $0.827 \pm 0.064\%$ result is then honestly compared, rather than equated, with Allart et al.'s $1.08 \pm 0.05\%$ transit-model measurement over a $0.75\text{ }\text{\AA}$ passband. Published follow-up literature quotes the same Allart result and individual-night depths. ² The 12.9σ number should remain labelled exactly as it is now: a weighted-band S/N under independent-point assumptions, not a covariance-aware atmospheric escape detection significance. Code quality is good for a one-file analysis. A real next step would be fitting a transit model and correlated residual process, not merely refining the mean estimator.

LHS 475b. The repository asks a scientifically meaningful question and gets the broad answer right: the spectrum is compatible with being featureless; hydrogen-rich and clear methane models are disfavoured; CO_2 -rich/high-cloud/no-atmosphere alternatives remain degenerate. This matches the published conclusion. ⁹ The code's flat fit uses $N - 1$ degrees of freedom appropriately because one mean level is fitted, while the candidate-model comparisons divide χ^2 by N . If the upstream model spectra were offset to the measured depth using an effectively fitted normalisation, then the appropriate effective degrees of freedom are not necessarily N ; this should be documented more carefully. More importantly, the README correctly admits that $\chi^2_\nu > 2$ is only a heuristic. I would now complete the job: print χ^2 , dof and survival-function p-values for every fixed model. That is only a few lines with `scipy.stats.chi2.sf` and immediately upgrades the statistical professionalism.

Transit-photometry method. The expanded search over period, phase and duration is a good injection-recovery exercise. The recovered period and depth demonstrate that the implementation can locate the planted signal without being handed most of the answer, and the README is unusually good at distinguishing the quoted depth S/N from a search-wide false-alarm probability. One factual claim should nevertheless be changed: BLS is described as the algorithm class used by "the Kepler and TESS pipelines". The official Kepler/TESS Transit Planet Search architecture uses an **adaptive, wavelet-based matched filter**, not simply BLS; BLS is a standard and widely used transit-search algorithm, but it is inaccurate to equate the custom search here with the Kepler/TESS pipeline core. The NASA pipeline literature describes the adaptive wavelet matched-filter approach for both missions. ¹⁸ The custom score is also BLS-like rather than a complete reproduction of the Kovács formal statistic. Finally, the worked HD 209458b arithmetic has tiny transcription errors: $15.58 \times 6371 = 99,260$ km, not 99,240, and $1.19 \times 695,700 = 827,883$ km, not 827,733; the final 14,375 ppm ratio is nevertheless correct.

Radial-velocity method. Of the four method demonstrations this is the one I would most readily show an astronomer. It solves Kepler's equation, implements the eccentric RV model correctly, uses irregular seasonal sampling, finds the initial period with Lomb-Scargle, performs a bounded nonlinear fit and

then inverts the semi-amplitude relation for $M_p \sin i$. The corrected $M_*^{-2/3}$ scaling is physically right. The simulation is deliberately easy— $K = 92 \text{ m s}^{-1}$ versus only $\sim 1.8 \text{ m s}^{-1}$ combined noise—so sub-percent recovery demonstrates numerical consistency rather than difficult detection performance. The README admits that. The 51 Peg worked check is also honest about its circularity: the archive's Rosenthal et al. solution gives $P = 4.2307969 \text{ d}$, $M_p \sin i = 0.464 M_J$, $e \approx 0.0042$ and $K = 55.73 \text{ m s}^{-1}$, so deriving the same K from a mass that was itself inferred from K is an internal-consistency check, not independent validation. ¹⁹ A minor code bug remains in the summary CSV: the “n_observations” recovered column uses the size of the seasonal candidate pool rather than the final selected data-vector length. Add a jitter parameter, FAP, posterior sampler and low- K injection suite and this becomes a very respectable methods exercise.

Gravitational-microlensing method. The blind anomaly-location fix is real, but this remains the scientifically weakest of the four simulations. The source light curve is a legitimate PSPL event, yet the “planetary” signal is a Gaussian whose amplitude and width are explicitly generated as fixed coefficients times $\sqrt{q}$; the recovery then fits a Gaussian and **inverts those same coefficient relations**. Consequently, the impressive $\sim 1\%$ mass-ratio recovery mostly establishes that the inverse of the injection rule recovers the injected number. It is not evidence that the pipeline could infer q from a real binary-lens caustic. Moreover, the amplitude- and duration-derived q values are described as two “independent” estimates, but they come from the same anomaly fit and are not statistically independent. The OGLE-2005-BLG-390Lb worked calculation itself is sound: the observed lens mass ratio is $q = 7.6 \pm 0.7 \times 10^{-5}$, while the quoted $\sim 5.5 M_{\oplus}$ planet and $\sim 0.22 M_{\odot}$ host have large physical-mass uncertainties. ²⁰ One current factual item also needs immediate updating: the README says Roman's launch is expected in 2027, whereas NASA currently lists **30 August 2026** as the planned launch date. ²¹

Direct-imaging method. The conceptual ADI sequence is well designed for teaching: static instrument-frame speckles, sky rotation of the companion, reference subtraction, de-rotation and coherent combination make the essential geometry immediately understandable. The companion's flux is now normalised against the star's aperture flux, which fixes an important dimensional problem. But I would not call the resulting curve “properly small-sample calibrated” yet. The code requests 12 azimuthal apertures irrespective of separation and then excludes some near the companion. Independent resolution elements should scale approximately with annular circumference divided by the PSF resolution element; near the star there are fewer and farther out there are more. Mawet et al.'s result is specifically about this small- N , separation-dependent regime. ⁶ The code also uses the population standard deviation (`np.std`, `ddof=0`) while invoking a finite-sample t -framework. A publication-grade contrast curve additionally requires fake-planet throughput calibration across separation and position angle, which the README commendably already lists as a limitation. The HR 8799 example is numerically sensible and is discussed below.

Worked cross-checks against real systems

These four README “real target” examples are useful because they reveal whether the pedagogical formulae are being used honestly.

Example	Independent check	Verdict
Transit — HD 209458b	With the README's $R_p = 15.58 R_\oplus$ and $R_\star = 1.19 R_\odot$, $R_p = 99,260$ km and $R_\star = 827,883$ km, giving $(R_p/R_\star)^2 = 0.014375$, or 14,375 ppm . Early observed transits were indeed at roughly the 1.5–1.7% scale and required limb-darkened transit modelling. ²²	Final depth arithmetic correct; two printed intermediate multiplications are slightly wrong. The comparison to a modern wavelength-dependent atmospheric depth should be described as scale agreement, not validation of BLS.
RV — 51 Peg b	The NASA Exoplanet Archive's Rosenthal et al. solution lists $P = 4.2307969$ d, $M_p \sin i = 0.464 \pm 0.014 M_J$, $e = 0.0042^{+0.0046}_{-0.0030}$, and $K = 55.73^{+0.32}_{-0.30}$ m s ⁻¹ . ¹⁹ Substitution of the README values gives about 55.74 m s⁻¹ .	Numerically correct but explicitly circular , because $M_p \sin i$ and K belong to the same orbital solution. Good as dimensional/internal consistency only.
Microlensing — OGLE-2005-BLG-390Lb	$5.5 M_\oplus / (0.22 M_\odot) \approx 7.5 \times 10^{-5}$, in excellent agreement with the directly fitted microlensing mass ratio $q = 7.6 \pm 0.7 \times 10^{-5}$. The physical masses themselves are much less precise: roughly $5.5^{+5.5}_{-2.7} M_\oplus$ and $0.22^{+0.21}_{-0.11} M_\odot$. ²⁰	Good order-of-magnitude check , but the direct q measurement is actually the cleaner quantity and should be shown first.
Direct imaging — HR 8799b	$1.713'' \times 39.4 \text{ pc} \approx \mathbf{67.5 \text{ AU}}$, consistent with the ~68 AU projected/orbital scale reported for HR 8799b. ²³ For a 10-m telescope at $1.6 \mu\text{m}$, $1.22\lambda/D = 0.0403''$, so $1.713''$ is ~42.5 times that nominal scale.	Arithmetic sound. Saying angular diffraction resolution was “never” a bottleneck is rhetorically too absolute; coronagraphic inner working angle, residual stellar PSF and contrast still govern detectability, although HR 8799b is indeed far outside one λ/D .

These examples are worth retaining. They demonstrate that the author understands how the pedagogical equations connect to physical systems. I would simply relabel them “**sanity checks against published system parameters**” rather than “validation”, because only an independent observable or independent analysis can really validate a method.

Ranked red flags and priority improvements

Specific errors and red flags, ranked by severity

Rank	Severity	Repository	Problem	Why a specialist will care
1	High	GJ 1214b	Low/negative nightside flux lower bounds are still replaced by 10^{-12} before Planck inversion.	Nonlinear transforms of low-S/N fluxes require posterior propagation. Hard clipping biases temperature intervals and can make weak measurements appear better constrained than they are.
2	High	WASP-121b	Hundreds of wavelength-dependent brightness temperatures are inverse-variance averaged into a scalar with a few-kelvin formal error.	A monochromatic brightness temperature is not a repeated measurement of one wavelength-independent thermodynamic parameter. Published effective temperatures require spectral/bolometric modelling. ²⁴
3	High-moderate	Direct imaging	“Independent sample” count for Mawet correction is effectively fixed by a 12-aperture grid instead of being derived from separation/resolution-element packing.	This undermines the statistical calibration of the claimed 5σ contrast curve—the very feature advertised as an advance over a toy ADI demonstration.
4	Moderate, but embarrassing	TRAPPIST-1e	Webpage labels NIRSpec PRISM $R \sim 10,000$.	Official JWST documentation gives roughly $R \sim 100$. ²⁵ An instrument specialist will notice immediately.
5	Moderate	HD 209458b	Code still calls four correlated tree leaves a genuine systematic-uncertainty estimate despite README caveats.	Internal contradiction suggests the scientific interpretation was fixed in prose but not fully fixed in the analysis.

Rank	Severity	Repository	Problem	Why a specialist will care
6	Moderate	Transit method	README states BLS is effectively the algorithm used by the Kepler/TESS pipelines.	The mission Transit Planet Search uses an adaptive wavelet matched-filter architecture; BLS is related in purpose, not the same pipeline. ¹⁸
7	Moderate	Microlensing	q is injected through Gaussian amplitude/width $\propto \sqrt{q}$, then recovered by fitting the same Gaussian form and algebraically inverting those same scalings.	The near-perfect recovery is substantially model-matched and should not be presented as evidence of realistic binary-lens inference.
8	Moderate	Microlensing	README calls amplitude- and duration-derived q values “independent”; Roman launch date says 2027.	The estimates share the same anomaly fit, and the launch date is stale: NASA currently lists 30 August 2026. ²¹
9	Moderate	55 Cnc e	Difference between local and Demory brightness temperatures partly blamed on reflected light.	The source paper documents use of an observed stellar IR spectrum; reflected light is not the documented explanation in that calculation. ¹⁷
10	Moderate-low	LHS 475b	Candidate models get χ^2_ν using N dof even though model vertical offsets were applied upstream; no p-values are calculated.	Not fatal, especially because the heuristic is admitted, but easy to make statistically cleaner.
11	Low-moderate	HD 189733b	Local feature S/N assumes diagonal errors and fixed windows; shared continuum/window choice is not incorporated.	Fine as a visual/statistical diagnostic, not as discovery significance. The repo mostly says this correctly.
12	Low-moderate	WASP-96b	Limb-difference significance neglects covariance.	Scientifically important but explicitly disclosed, so this is now a limitation rather than an overclaim.

Rank	Severity	Repository	Problem	Why a specialist will care
13	Low	K2-18b	Hand-picked band statistic ignores covariance and is conditional on one particular reduction/offset treatment.	The repo already refuses to turn it into a retrieval claim, which makes this acceptable.
14	Low	WASP-107b	Peak-to-trough spectrum amplitude lacks a statistical uncertainty or null distribution.	Descriptive only; do not imply it independently establishes molecular absorption.
15	Low	HAT-P-11b	12.9σ assumes phase-folded data points are independent and uses an arbitrary fixed phase cut.	Proper transit/GP fitting would change the formal uncertainty, but the limitation is clearly disclosed.
16	Low	Radial velocity	Summary-CSV observation-count field does not report the actual selected data-vector size; no FAP/jitter inference.	Minor code-quality issue plus known pedagogical simplification; not a physics error.

Highest-impact improvements

Priority	Change	Scientific payoff
1	Adopt posterior/covariance-aware uncertainty propagation wherever the source release permits it. For Planck inversions, propagate posterior draws rather than $x \pm \sigma$; for spectra, use covariance matrices or explicitly state “diagonal likelihood”.	This fixes the largest gap between the portfolio's polished appearance and research-grade inference. GJ 1214b should be the first repair.
2	Replace scalar weighted averages of wavelength-dependent brightness temperature with physically defined quantities. For WASP-121b and GJ 1214b, either leave $T_b(\lambda)$ as a spectrum or derive an effective temperature through a band-integrated/bolometric emission model.	Prevents formally spectacular but physically misleading $\pm$ few-K headline uncertainties.
3	Fix the visible specialist errors immediately: TRAPPIST-1e $R \sim 100$, not 10,000; Roman launch date; Kepler/TESS BLS wording; 55 Cnc reflected-light explanation; HD 209458b “independent pipeline/systematic uncertainty” language. ²⁶	These are cheap fixes and disproportionately affect first impressions.

Priority	Change	Scientific payoff
4	Add automated scientific tests. Examples: Planck inversion round-trip tests; Kepler equation residual $< 10^{-10}$; synthetic BLS period recovery across several seeds; RV mass-amplitude round trips; χ^2 values on a frozen mini-dataset; ADI rotation sign and flux-throughput tests.	A working astronomer is much more likely to trust code whose physics is encoded as executable assertions.
5	Add CI and a truly reproducible environment. Use a lockfile/conda environment or pinned tested versions, GitHub Actions to run analyses/tests, and record Python/package versions in generated metadata.	Moves the portfolio from “scripts that probably rerun” toward reproducible computational research.
6	For the direct-imaging repository, rebuild the contrast curve properly: define independent resolution elements from FWHM/ λ/D , use sample variance, inject fake planets over separation/PA, measure algorithmic throughput and apply small-sample false-positive calibration.	Would turn that repo from a visually good ADI demo into a surprisingly serious high-contrast-imaging exercise. Mawet et al. directly motivate this. ⁶
7	For microlensing, replace the Gaussian planetary bump with a real binary-lens calculation. Using <code>MuLensModel</code> as a benchmark is fine even if a from-scratch binary-lens solver is too large; inject with one implementation and recover with another.	This is the single biggest opportunity to eliminate the “toy/model-matched” impression among the method repos.
8	For RV and transit, add difficult injection suites rather than one spectacular recovery. Sweep K/depth, red-noise amplitude, cadence/gaps and stellar variability; use noise-only simulations to estimate empirical false-alarm/completeness curves.	Detection efficiency versus S/N is much more scientifically informative than recovering one deliberately obvious signal at sub-percent precision.
9	Upgrade χ^2 repositories from heuristics to explicit hypothesis tests. Report χ^2 , dof, p-value, assumptions, fitted nuisance parameters and, where models are non-nested, preferably information criteria or Bayesian evidence only when justified.	LHS 475b and TRAPPIST-1e could gain substantial statistical credibility with very little extra code.
10	Create one portfolio-level methodology/provenance page that labels each repo by analysis level: raw/reduced data, published secondary products, independent statistic, literature retrieval result, synthetic pedagogical demonstration.	Prevents an admissions reviewer from mistaking “real archival data” for a claim of original detector-level reduction and makes the author’s actual competence easier to evaluate fairly.

The single most important scientific improvement is **not adding another repository**. It is making uncertainty treatment commensurate with the sophistication of the presentation. At present the webpages can visually resemble mini-papers while several headline error bars still come from diagonal, independent-bin approximations. A portfolio becomes much more impressive when it says, in effect, “here is the covariance-aware answer; here is what the naïve independent-bin answer would have been; here is why they differ.”

Final verdict

Overall grade: B+ — a genuinely credible and unusually thoughtful independent exoplanet/data-analysis portfolio, clearly above superficial plotting work, but not “publishable-quality analysis” as a body because covariance/posterior treatment, physical interpretation of some aggregate statistics, simulation realism and software validation still lag behind the presentation.

Nothing in the portfolio makes the author look scientifically unserious. Several features do the opposite: the explicit separation of local statistics from published retrievals, use of traceable real data products, correction of earlier errors and willingness to state limitations are all strong signals of research maturity. But **I would not send the portfolio unchanged to a specialist today**. The TRAPPIST-1e $R \sim 10,000$ claim, the residual GJ 1214b flux clipping, the WASP-121b few-kelvin cross-wavelength average, the HD 209458b “genuine systematic uncertainty” wording, the Kepler/TESS=BLS statement, and the direct-imaging small-sample implementation are exactly the sort of details a domain expert may use to decide whether the apparent rigour is real. Correct those, add executable tests, and reframe the analyses consistently as **independent quantitative checks on published reduced data rather than miniature publication-grade retrievals**, and I would regard the portfolio as a strong asset for graduate/RA applications rather than something needing explanation or apology.

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